PROGRAM-7

1. Input Details
For each product:

- **Product Name**
- **Category**
- **Unit Price**
- **Quantity**

2. Item Amount Calculation

$$\text{Item Amount} = \text{Unit Price} \times \text{Quantity}$$

$$\text{Subtotal} = \sum \text{Item Amounts}$$

3. Category-Specific Taxes
Example tax rates:

Category	Tax Rate
Grocery	5%
Electronics	18%
Clothing	12%
Stationery	8%

Tax Calculation

$$\text{Tax} = \text{Item Amount} \times \frac{\text{Tax Rate}}{100}$$

4. Threshold Discounts
Example discount rules:

Purchase Amount	Discount
₹5,000 - ₹9,999	5%
₹10,000 - ₹19,999	10%
Above ₹20,000	15%

$$\text{Discount} = \text{Subtotal} \times \frac{\text{Discount Rate}}{100}$$

5. Maximum Concession Limit
To prevent excessive discounts:

$$\text{Maximum Discount Allowed} = ₹3,000$$

If calculated discount exceeds ₹3,000:

$$\textbf{Discount} = \textbf{₹3,000}$$

6. Final Bill Calculation

$$\text{Total Tax} = \sum \text{Product Taxes}$$
$$\text{Net Amount} = \text{Subtotal} + \text{Total Tax} - \text{Discount}$$

7. Algorithm

1. Input product details.
2. Calculate item amount for each product.
3. Compute tax according to product category.
4. Calculate subtotal.
5. Apply threshold discount.
6. Check maximum concession limit.
7. Calculate final payable amount.
8. Generate itemized invoice.
9. Display bill summary.

Example

Products Purchased

Product	Category	Price	Qty	Amount
Rice	Grocery	₹100	20	₹2,000
Shirt	Clothing	₹800	5	₹4,000
Headphones	Electronics	₹2,500	2	₹5,000

Subtotal

$$₹2{,}000 + ₹4{,}000 + ₹5{,}000 = ₹11{,}000$$

Tax

- **Grocery Tax = ₹100**
- **Clothing Tax = ₹480**
- **Electronics Tax = ₹900**

Total Tax

$$\textbf{₹100} + \textbf{₹480} + \textbf{₹900} = \textbf{₹1,480}$$

Discount

- **Purchase Amount = ₹11,000**
- **Eligible Discount = 10%**

$$\textbf{₹11,000} \times \textbf{10\%} = \textbf{₹1,100}$$

Since ₹1,100 is below the maximum concession limit, it is applied fully.

Final Bill

$$\textbf{₹11,000} + \textbf{₹1,480} - \textbf{₹1,100} = \textbf{₹11,380}$$

Itemized Invoice
Plain Text

```
----------------------------------------
RETAIL INVOICE
----------------------------------------
Product Qty Amount
Rice 20 ₹2,000
Shirt 5 ₹4,000
Headphones 2 ₹5,000
----------------------------------------
Subtotal ₹11,000
Total Tax ₹ 1,480
Discount ₹ 1,100
----------------------------------------
Net Amount ₹11,380
```

Programm

```
# Retail Billing with Multiple Discount Rules

n = int(input("Enter number of products: "))

items = []
subtotal = 0
total_tax = 0

for i in range(n):
    print("\nProduct", i + 1)

    name = input("Enter product name: ")
    category = input("Enter category (food/clothing/electronics): ").lower()
    price = float(input("Enter price per item: "))
    quantity = int(input("Enter quantity: "))

    # Category-specific tax rates
    if category == "food":
        tax_rate = 0.05
    elif category == "clothing":
        tax_rate = 0.08
    elif category == "electronics":
        tax_rate = 0.18
    else:
        print("Invalid category. Using 5% tax.")
        tax_rate = 0.05

    amount = price * quantity
    tax = amount * tax_rate

    subtotal += amount
    total_tax += tax
```

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    items.append((name, category, price, quantity, amount, tax))

# Calculate threshold discount
if subtotal >= 5000:
    discount_rate = 0.15
elif subtotal >= 3000:
    discount_rate = 0.10
elif subtotal >= 1000:
    discount_rate = 0.05
else:
    discount_rate = 0

discount = subtotal * discount_rate

# Maximum concession limit
max_discount = 1000

if discount > max_discount:
    discount = max_discount

# Calculate final bill
final_bill = subtotal + total_tax - discount

# Display itemized invoice
print("\n============= RETAIL INVOICE =============")
print(f"{'Product':<15}{'Qty':<6}{'Price':<10}{'Amount':<10}{'Tax':<10}")
print("-" * 51)

for item in items:
    name, category, price, quantity, amount, tax = item
    print(f"{name:<15}{quantity:<6}{price:<10.2f}{amount:<10.2f}{tax:<10.2f}")

print("-" * 51)
print(f"Subtotal:             Rs. {subtotal:.2f}")
print(f"Total Tax:            Rs. {total_tax:.2f}")
print(f"Discount:              Rs. {discount:.2f}")
print(f"Final Bill Amount:    Rs. {final_bill:.2f}")
print("=" * 51)
```

output:

```
Enter number of products: 9

Product 1
Enter product name: lakme
Enter category (food/clothing/electronics): clothing
Enter price per item: 78
Enter quantity: 65

Product 2
Enter product name: satin
Enter category (food/clothing/electronics): clothing
Enter price per item: 95
Enter quantity: 100
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Product 3
Enter product name: fired rice
Enter category (food/clothing/electronics): food
Enter price per item: 80
Enter quantity: 78